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MILITARY SPECIFICATION

# COVERING, ORGANIC PLASTIC (FOR INSTRUMENT BODIES)

## 1. SCOPE

**1.1** This specification covers organic plastic covering for instrument bodies.

## 2. APPLICABLE DOCUMENTS

**2.1** The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

### SPECIFICATIONS

FEDERAL

L–P–406 — Plastics, Organic: General Specification, Test Methods.
UU–P–268 — Paper, Kraft, Wrapping.
UU–T–111 — Tape; Paper, Gummed (Sealing and Securing).
PPP–B–585 — Boxes, Wood, Wirebound.
PPP–B–591 — Boxes, Fiberboard, Wood-Cleated.
PPP–B–601 — Boxes, Wood, Cleated-Plywood.
PPP–B–621 — Boxes, Wood, Nailed and Lock-Corner.
PPP–B–636 — Box, Fiberboard.
PPP–T–60 — Tape, Pressure Sensitive Adhesive, Waterproof —for Packaging and Sealing.
PPP–T–76 — Tape, Pressure-Sensitive Adhesive, Paper, Water Resistant.
PPP–T–97 — Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

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MIL–B–10377 — Box, Wood, Cleated, Veneer, Paper Overlaid.
MIL–L–10547 — Liners, Case, Waterproof.

### STANDARDS

MILITARY

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–129 — Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

**2.2 Other publications.** The following documents forms a part of this specification to the

FSC 6650

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extent specified herein. Unless otherwise indicated, the issue on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.

## 3. REQUIREMENTS

**3.1 Material.** The organic plastic shall consist of an acrylonitrile butadiene styrene copolymer resin with or without the addition of compounding ingredients such as rubber modifiers, dyes, pigments, fillers, and stabilizers.

**3.2 Preparation of metal to receive covering.** Aluminum instrument bodies shall be anodized before the covering is applied.

**3.3 Accelerated light and weather aging.** When subjected to the accelerated light and weather aging test specified in 4.4.3, the covering on the instrument body shall show no signs of deterioration.

**3.4 Abrasion.** When tested with the wear-test machine as specified in 4.4.4, the embossing shall not be worn completely away by 500 revolutions of the abrasion disks. The average decrease in thickness after 2,400 revolutions shall be not more than 0.005 inch.

**3.5 Salt spray.** When subjected to the salt spray test specified in 4.4.5, there shall be no separation of the covering material from the instrument body, opening at the seams, nor deterioration of the surface of the plastic covering.

**3.6 Resistance to impact.** When subjected to the impact test specified in 4.4.1, the covering shall not chip or crack, nor shall any permanent impression be left in the covering.

**3.7 Adhesion of covering to instrument body.** The plastic covering shall adhere firmly and uniformly to the instrument body and shall have suitably sealed seams and edges. When tested as described in 4.4.2, the average adhesion of the covering to the body shall be not less than 1.0 pounds per ¼ inch of width of covering. The adhesion shall be at least 0.5 pound per ¼ inch of width at every point tested.

**3.8 Thickness.** The thickness of the plastic sheeting shall be 0.018 to 0.030 inch.

**3.9 Surface finish.** The covering shall be black and uniformly embossed to provide a dull grained surface.

**3.10 Uniformity.** All sheets shall be uniform in texture, finish, and specified properties.

**3.11 Surface defects.** The material shall be free from blisters, wrinkles, or cracks and reasonably free from minor defects such as scratches, dents, and heat marks.

**3.12 Dimensions and tolerances.** Unless otherwise specified (see 6.2), the manufacturer's standard sizes between 30 and 52 inches in width and between 36 and 120 inches in length will be acceptable. The length and width of sheets may vary ⅛ inch over or under the manufacturer's standard size.

**3.13 Workmanship.** The workmanship shall be first class in every respect.

## 4. QUALITY ASSURANCE PROVISIONS

**4.1** The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

**4.2 Sampling.**

**4.2.1** *Inspection lot.* All plastic, in sheet form, produced in one manufacturing batch, and offered for delivery at one time, shall be considered a lot for purposes of sampling and inspection.



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**4.2.2** *Sampling for visual examination.* Sample sheets shall be selected at random from each lot in accordance with Standard MIL–STD–105 at inspection level III for the examination of 4.3. At least half of each sheet shall be examined. The Acceptable Quality Level (AQL) shall be 1.5 percent defective.

**4.2.3** *Sampling for group B tests.* Sample sheets shall be selected in accordance with inspection level L–5 of table for small sample inspection of Standard MIL–STD–105, and sufficient material shall be cut from each sample sheet to provide a specimen for each test in 4.4.1 and 4.4.2. If any specimen fails the tests specified in 4.4.1 or 4.4.2, the lot represented by the sample shall be rejected.

**4.2.4** *Sampling for group C tests.* Sample sheets shall be selected in accordance with inspection level L–3 of the table for small sample inspection of Standard MIL–STD–105, and sufficient material shall be cut from each sheet to provide one specimen for each test of 4.4.3, 4.4.4, and 4.4.5. If any specimen fails the tests specified in 4.4.3, 4.4.4, and 4.4.5, the lot represented by the sample shall be rejected.

**4.3 Visual and dimensional examination.** Each of the sample sheets selected in accordance with 4.2.2 shall be visually and dimensionally examined to verify compliance with this specification. Any sheet in the sample containing one or more visual defects shall be rejected, and if the number of defective sheets in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

**4.4 Test procedures.**

**4.4.1** *Impact test.* The covered brass or aluminum tubing shall be conditioned for 1 hour at a temperature of —20° Fahrenheit (F.). Within 1 minute after removal from the cooling medium, the specimen shall be clamped securely on a solid table and a 1-pound steel ball shall be allowed to make a free fall of 24 inches onto the instrument cover. The ball shall strike with a normal impact, that is, the ball shall rebound in an approximately vertical direction.

**4.4.2** *Adhesion.* The specimens shall be prepared by scoring the covering on the brass or aluminum tubing at 2 representative locations and then removing ⅛-inch wide strips approximately 2 inches long and ¼ inch apart, in order to leave three ¼-inch wide strips which run circumferentially. The ends of these strips shall be lifted with a sharp instrument so as to detach them for a distance of approximately ⅜ inch. A clamp shall be securely fastened to the ends of the strips. A small bucket for steel shot shall be suspended from the clamp (weight not to exceed 1/10 of a pound). The steel shot shall be added at the approximate rate of 1 pound per minute. The adhesion shall be determined by measuring the weight of the steel shot required to cause failure of the bond or of the strip of covering material. The test shall be conducted at 77° + 2° F. and 50 percent relative humidity. The brass or aluminum tubing shall be so supported and manipulated that the vertically downward line of load application shall be perpendicular to the surface of the body whether curved or flat. An average of 5 readings shall be made on each strip, making a total of fifteen determinations for each brass or aluminum tube.

**4.4.3** *Accelerated light and weather aging.* The accelerated light and weather aging test shall be conducted as described in method No. 6022 of Specification L–P–406, except that the unit shall be operated with "Sunshine" carbons and without filters. The covered brass or aluminum tubing shall be so fastened that the area containing a seam shall be directly exposed to the light source, and the center of the specimen shall be at the same relative height as the center of the arc. The specimen shall be exposed for 100 hours.

**4.4.4** *Abrasion test.*

**4.4.4.1** *Specimens.* The specimens shall consist of three pieces of the covering material 2 by 3 inches which have been subjected to the accelerated light and weather aging test described in 4.4.3.

**4.4.4.2** *Apparatus.* Resistance to abrasion shall be determined using a wear-test machine having the following essential mechanical char-



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acteristics as designated on and illustrated by figure 1:

(a) Mild steel abrasion disk, 14⅛ inches in diameter and ½-inch thick, rotated clockwise at a constant speed of 23.5 revolutions per minute (r.p.m.).

(b) Specimen holder, 2 by 4 inches, attached to a ⅝-inch diameter shaft by means of a universal coupling. The distance between centers of the disk and the specimen holder shaft is 5.1 inches. The specimen shall be cemented to a suitable backing and secured to the specimen holder by means of clamps at each end of the holder. The holder revolves clockwise at a constant speed of 32.5 r.p.m.

(c) Hardened tool steel cam attached to the specimen holder shaft by means of set screws. In each one-half revolution of the shaft the specimen is lifted a distance of one-sixteenth inch and then dropped back on disk (described in (a) above). Change in the lift due to wear of the specimen is compensated by adjustment of the clam at regular intervals so that the lift is maintained practically constant.

(d) Weight to provide a total load of 10 pounds acting on the specimen.

(e) Distributor for abrasive grits. The grits flow by gravity from the hopper (f) through a 13/64-inch inside diameter tube and into the center of the distribution chamber. An impeller in the chamber provides an evenly distributed feed of the grits through eight 3/64-inch diameter holes centered on a 1⅝-inch diameter pitch circle. A uniform distribution of grits falls to disk (a) in the path of the test specimen. The distributor is rotated at a constant speed of 9 r.p.m.

(f) Hopper for feeding abrasive grits.

(g) V-belt connected to a reduction gear driven by a constant speed ⅓-horsepower motor.

(h) Revolution counter (for counting revolutions of the abrasion disk).

(i) Wooden frame.

(j) Hopper for collecting used abrasive grits.

All rotating parts of the machines are driven by gears or roller chains.

**4.4.4.3** *Procedure.* The specimen shall be cemented to a suitable backing and secured to the specimen holder. Before starting the test, the thickness of the specimen shall be measured at 12 equally distributed points by means of a dial thickness gage and a template. During the test, a loose standard abrasive consisting of No. 80 aluminum oxide shall be continuously fed at the rate of approximately 44 grams per minute to the revolving disk. After 500 revolutions of the abrasive disk, the surface of the specimen shall be examined to determine whether any of the original embossed surface remains. The test shall then be continued until 2,400 revolutions have been completed and the thickness of the specimen shall be determined again at the points originally measured. The average of the differences between the original and the final readings shall be taken as the average decrease in thickness of the covering material after 2,400 revolutions.

**4.4.5** *Salt spray test.*

**4.4.5.1** A suitable apparatus for performing the salt spray test is illustrated by figure 2. Deviations from this apparatus are permissible but the apparatus must embody the essential features shown on figure 2. The capacity of the chamber shall be approximately 6 cubic feet. The spray nozzle shall be supplied with a volume of air mounting to five times the capacity of the chamber each hour. The air shall be supplied to the nozzle free from oil and dirt and at a pressure of 10 to 15 pounds per square inch and a relative humidity of not less than 90 percent. The nozzle shall fill the chamber with a uniform fog or mist, and shall not impinge directly on the specimens. All parts of the equipment shall be of corrosion-resisting material. When in operation, a 20-percent solution of sodium chloride in water



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at a pH of 6 to 7 shall be maintained in the reservoir feeding the nozzle. Suitable means shall be provided to maintain the specimen chamber at 90° ± 4° F.

**4.4.5.2** *Procedure.* The covered instrument shall be suspended in the specimen chamber and subjected to a continuous salt-water spray for 96 hours. Temperature measured near the center of the chamber shall be recorded every 12 hours. The concentration and pH of the salt solution in the reservoir shall be measured and corrected at least once every 24 hours.

**4.4.6** *Preparation of specimens for test.* All tests except the resistance to abrasion (see 3.4) shall be conducted on sheet material bonded to seamless brass tubing (1.660 inches o.d.) and aluminum tubing (1.50 inches o.d.) which are considered to be representative of the bodies of portable navigational instruments. Adhesives shall be applied in accordance with manufacturer's instructions and allowed to cure a minimum of 96 hours at 70° ± 5° F. and 50 ± 2 percent relative humidity before being subjected to tests.

## 5. PREPARATION FOR DELIVERY

**5.1 Packaging.** Packaging shall be level A or C as specified (see 6.2).

**5.1.1** *Level A.* Material shall be wrapped with kraft paper conforming to Specification UU–P–268. All seams, joints and closure shall be sealed with adhesive conforming to Specification MIL–A–140 or tape conforming to Specification UU–T–111.

**5.1.2** *Level C.* Material shall be packaged in accordance with the supplier's commercial practice.

**5.2 Packing.** Packing shall be level A, B, or C as specified (see 6.2).

**5.2.1** *Level A.* Material packaged as specified (see 6.2) shall be packed in containers conforming to any one of the following specifications at the option of the supplier:

| *Specification* | *Type or class* |
|---|---|
| PPP–B–585 | Class 3 use |
| PPP–B–591 | Overseas type |
| PPP–B–601 | Overseas type |
| PPP–B–621 | Class 2 |
| PPP–B–636 | Class 2 |
| MIL–B–10377 | Overseas type |

Shipping containers shall have caseliners conforming to Specifications MIL–L–10547. Caseliners shall be closed and sealed in accordance with the appendix to Specification MIL–L–10547. Caseliners for class 2 fiberboard boxes conforming to Specification PPP–B–636 may be omitted provided all corners and edge seams and manufacturer's joints are sealed with minimum 2-inch wide tape conforming to Specification PPP–T–76 or PPP–T–60. Boxes shall be closed, strapped, or banded in accordance with the applicable box specification or appendix thereto. The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds.

**5.2.2** *Level B.* Material, packaged as specified (see 6.2), shall be packed in boxes conforming to any of the following specifications at the option of the supplier:

| *Specification* | *Type or class* |
|---|---|
| PPP–B–585 | Class 1 or 2 use |
| PPP–B–591 | Domestic type |
| PPP–B–601 | Domestic type |
| PPP–B–621 | Class 1 |
| PPP–B–636 | Class 1 |
| MIL–B–10377 | Domestic type |

**5.2.3** *Level C.* Material, packaged as specified (see 6.2), shall be packed in containers which will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the Uniform Freight Classification Rules or other regulations as applicable to the mode of transportation.

**5.3 Marking.** In addition to any special marking required by the contract or order or herein, interior and exterior shipping containers shall be marked in accordance with Standard MIL–STD–129.



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## 6. NOTES

**6.1 Intended use.** Organic plastic material is intended for use as a cover for main body tubes of binoculars, spyglasses, etc.

**6.2 Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) The instruments to be covered.

(c) The length and width of the sheet (see 3.12).

(d) Applicable levels of packaging and packing (see 5.1 and 5.2).

**Notice.** When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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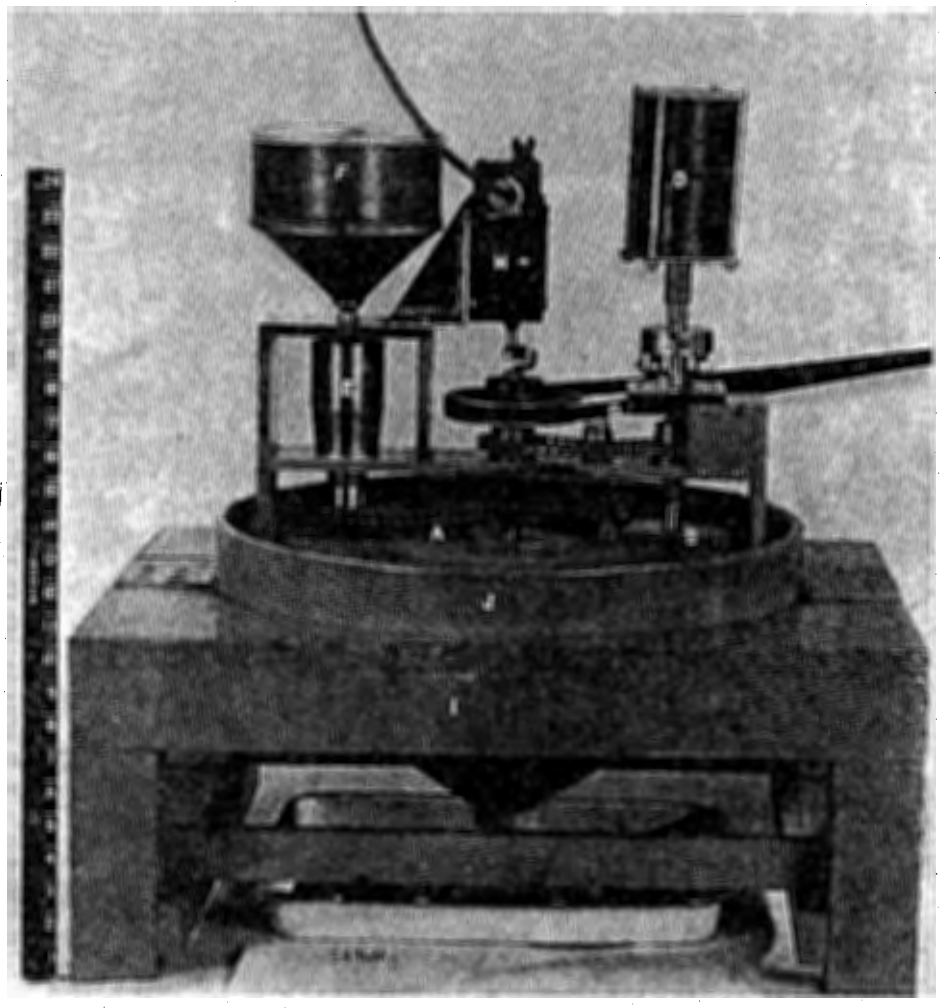


FIGURE 1. *Wear-test machine.*



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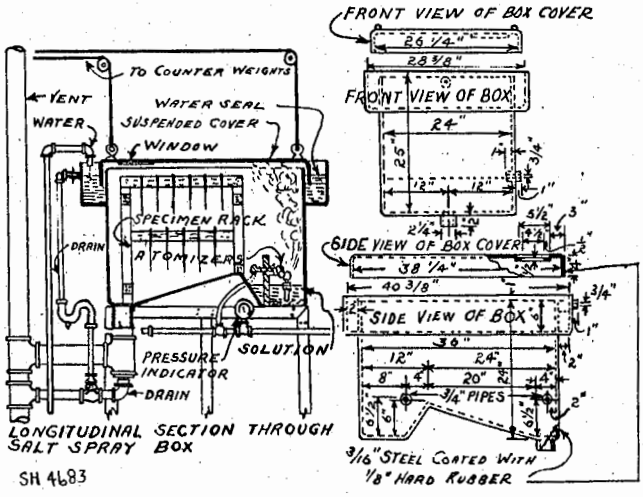



FIGURE 2. *Apparatus for salt spray test.*



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